

Last Time

Nonparametric Procedures

- Bootstrapping

- Permutation Tests

- Wilcoxon Rank-Sum Test

- Wilcoxon's Matched-Pairs Signed-Ranks Test

Today

Wilcoxon's Rank Tests

Wilcoxon's Matched-Pairs Signed-Ranks Test

χ^2 Tests of Goodness of Fit and Association

Goodness of Fit Example

Test of Association Example

Summary

Wilcoxon rank sum tests (Mann-Whitney U tests) are non-parametric alternatives to the Student's and Welch's t-tests.

- ▶ Do *not* assume the data are normally distributed, equal variance, etc.
- ▶ Robust to outliers and skewness, especially with small sample sizes or ordinal data
- ▶ $H_0 : F_x = F_Y$
 - ▶ Wilcoxon rank sum test: replaces the two independent sample t-tests
 - ▶ Wilcoxon signed rank test: replaces the dependent (paired) samples t-test
- ▶ Asymptotically equivalent to t-tests
- ▶ Dealing with tied ranks

Matched Pairs

- ▶ While the rank sum test gives the nonparametric analog to the two independent samples t -test,
- ▶ the Wilcoxon signed-ranks test gives the nonparametric analog to the two dependent samples t -test.
- ▶ Rank the absolute differences for each group.

Suddath et al. 1990

Pair	Normal	Schizophrenic	Difference	Rank	Signed Rank
1	1.94	1.27	0.67	15	15
2	1.44	1.63	-0.18	9	-9
3	1.56	1.47	0.09	5	5
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
15	2.08	1.97	0.11	7	7

Let R_i be the signed rank for the i^{th} set of twins. Then the test statistic is:

$$T = \min \left\{ \left| \sum_{R_i > 0} R_i \right|, \left| \sum_{R_i < 0} R_i \right| \right\}$$

- ▶ Let $T^+ = \sum_{R_i > 0} R_i = 15 + 5 + 10 + \dots = 111$.
- ▶ Let $T^- = \sum_{R_i < 0} R_i = -9$.
- ▶ Then $T = \min\{|111|, |-9|\} = 9$.

In R

```
> # Use the schizophrenia volume numbers
> twin.1 <- c(1.94,1.44,1.56,1.58,2.06,1.66,1.75,
>            1.77,1.78,1.92,1.25,1.93,2.04,1.62,
>            2.08)
> twin.s <- c(1.27,1.63,1.47,1.39,1.93,1.26,1.71,
>            1.67,1.28,1.85,1.02,1.34,2.02,1.59,
>            1.97)
>
> wilcox.test(x=twin.s,y=twin.1,alternative="less",
>            paired=TRUE,exact=TRUE,correct=FALSE)
```

Wilcoxon signed rank exact test

```
data: twin.s and twin.1
V = 9, p-value = 0.001007
alternative hypothesis: true location shift is less than 0
```

Ties

- ▶ Define N = number of pairs (individuals).
- ▶ If $X_i - Y_i = 0$, eliminate the pair and reduce N by 1.
- ▶ If ranks are tied,
 - ▶ and both ranks have the same sign, break the tie however you want.
 - ▶ and both ranks have a different sign, assign tied ranks and proceed as usual.

Normal Approximation

- ▶ When N_1, N_2 get very large, the distribution of W_S is approximately normal. Ordering $N_1 + N_2$ observations becomes very difficult.
- ▶ Because

$$\begin{aligned}E(W_S) &= \mu_{W_S} = \frac{N_1(N_1 + N_2 + 1)}{2} \\ \text{Var}(W_S) &= \sigma_{W_S}^2 = \frac{2N_2\mu_{W_S}}{12} = \frac{N_1N_2(N_1 + N_2 + 1)}{12}.\end{aligned}$$

- ▶ So,

$$z_{W_S} = \frac{W_S - \mu_{W_S}}{\sigma_{W_S}} = \frac{W_S - \mu_{W_S}}{\sqrt{\frac{2N_2\mu_{W_S}}{12}}}.$$

R does a normal approximation unless you tell it not to.

χ^2 Tests

We have encountered the χ^2 distribution before:

- ▶ The sum of k squared standard normal random variables Z_i follows a χ^2 distribution with k degrees of freedom.
- ▶ The statistic $\frac{(N-1)s^2}{\sigma^2}$ follows a χ^2 distribution with $N-1$ degrees of freedom.

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The χ^2 distribution is also the basis of a number of nonparametric tests of primarily categorical data.

- ▶ There are two kinds of χ^2 tests:
 - ▶ Goodness of Fit (frequency distribution)
 - ▶ Independence (contingency table)

Logic of the χ^2 Goodness-of-Fit Test

- ▶ In the χ^2 goodness-of-fit test, H_0 makes a statement about proportions in the population.

$$H_0 : \pi_1 = p_1, \pi_2 = p_2, \dots,$$

vs. the alternative H_1 : at least one of the population proportions π_i does not equal its hypothesized value p_i .

- ▶ Compare the sample (observed) frequencies to those predicted by H_0 .

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- ▶ When H_0 is false, the discrepancies will be large and so will the X^2 statistic.
- ▶ To test H_0 , we compare the X^2 value to the χ^2 distribution given H_0 is true.

Expected vs. Observed

The χ^2 statistic is based on frequency data: The number of counts in each category defined by a categorical variable.

Example 1: Frequency Distribution

Consider the racial demographics of the OSU student body.

OSU Minority Students (10-Year Trend)

	Black	Asian	Hispanic	Two or more	American Indian or Alaska Native	Native Hawaiian/PI	Total
$f_o(j)$	5,449	6,517	3,895	3,038	48	31	18,978

Example 2: Contingency Table

	HIV positive	HIV negative	Total
No IV Drug Use	4	73	77
IV Drug Use	15	33	48
Total	19	106	125

The Frequency Distribution: Observed vs. Expected Frequencies

According to the **US Census Bureau**, the minority profile of the state of Ohio (2023) is:

Ohio Minority Residents

	Black	Asian	Hispanic	Two or more	American Indian or Alaska Native	Native Hawaiian/PI	Total
p_j	0.134	0.028	0.048	0.027	0.003	0.001	0.241

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Was the distribution of underrepresented students at OSU what we might expect given the proportions of underrepresented groups living in the state of Ohio? The discrepancy between observed (OSU) and expected (US Census) frequencies can be expressed as a X^2 statistic.

The χ^2 Goodness-of-Fit Test

The *observed frequencies* f_o are given by the OSU profile:

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Expected Frequencies

There are a total of $\sum_j f_O(j) = 18,978 = N$ minority students at OSU.

The *expected frequency* $f_E(j)$ in each minority group should be $p'_j N$ if the OSU profile matches the Ohio profile:

Expected Frequencies							
	Black	Asian	Hispanic	Two or more	Native American	Native Hawaiian/PI	Total
p_j	0.134	0.028	0.048	0.027	0.003	0.001	0.241
$p'_j = p_j / \sum_{i=1}^6 p_i$	0.556	0.116	0.199	0.112	0.124	0.004	1.000
$f_E(j) = N p'_j$	10,552.08	2,204.92	3,779.85	2,126.17	236.24	78.75	18,978

Logic of the Test

Note that even if $H_0 : p_{\text{Black}} = 0.134, p_{\text{Asian}} = 0.028$, etc. were true, we would not expect $f_O = f_E$ exactly for all categories.

How much variation might we reasonably expect?

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How much variation might we reasonably expect?

The X^2 statistic gives us a measure of the discrepancy, and we can compute probabilities associated with large and small discrepancies using the χ^2 table.

$$\begin{aligned} X^2 &= \sum_{j=1}^k \frac{(f_O(j) - f_E(j))^2}{f_e(j)} \\ &\left(= \sum_{j=1}^k \frac{(O_j - E_j)^2}{E_j} \right) \end{aligned}$$

This statistic follows a χ^2 distribution with $k - 1$ degrees of freedom.

Characteristics of X^2

$$X^2 = \sum_{j=1}^k \frac{(f_o(j) - f_e(j))^2}{f_e(j)}$$

- ▶ It cannot be negative,
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- ▶ Each discrepancy contributes to X^2 according to its size *relative to* the expected frequency,
- ▶ The more groups we have, the larger we expect X^2 to be (the mean of a $\chi^2(k)$ variable is k).

Example: Is minority recruitment at OSU effective?

Does the population of OSU undergraduates accurately reflect the ethnic profile of the Ohio population?

Expected vs. Observed Frequencies

	Black	Asian	Hispanic	Two or more	Native American	Native Hawaiian/PI	Total
f_o	5,449	6,517	3,895	3,038	48	31	18,978
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$$\begin{aligned}X^2 &= \sum_{j=1}^5 \frac{(f_o(j) - f_E(j))^2}{f_E(j)} \\&= 11,474.43\end{aligned}$$

Reject or fail to reject H_0 ? ($\alpha = .05$)

In R

```
> # OSU vs. State of Ohio demographics (2014)
> osu <- c(5449,6517,3895,3038,48,31)
> ohio <- c(.134,.028,.048,.027,.003,.001)

> # By hand
> obs <- osu
> # Compute expected frequencies
> expect <- (ohio/sum(ohio))*sum(osu) # Note normalization of probabilities
> rbind(obs,expect)

      [,1]      [,2]      [,3]      [,4]      [,5]      [,6]
obs    5449.00  6517.000  3895.000  3038.000   48.0000  31.00000
expect 10552.08  2204.913  3779.851  2126.166  236.2407  78.74689

> # Compute the chi-squared statistic and its p-value
> X2 <- sum((obs-expect)^2/expect)
> X2

[1] 11474.43

> 1-pchisq(X2,length(expect)-1)

[1] 0
```

In R

```
> # Using R  
> chisq.test(osu,p=ohio/sum(ohio))
```

Chi-squared test for given probabilities

```
data:  osu  
X-squared = 11474, df = 5, p-value < 2.2e-16
```

Assumptions

- ▶ Random sampling
- ▶ All observations are independent

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- ▶ Random sampling
- ▶ All observations are independent
- ▶ Categories are mutually exclusive
- ▶ $f_E(j) \geq 5$ for all j .

Contingency Table Example

Does IV drug use increase the rate of HIV infection in a population?

	HIV positive	HIV negative	Total
No IV Drug Use	4	73	77
IV Drug Use	15	33	48
Total	19	106	125

Probability Review

What does it mean for two variables to be independent?

$$P(A|B) = P(A)$$

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Mathematical definition of independence:

$$P(A \cap B) = P(A)P(B)$$

Determining Expected Frequencies

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$$P(HIV^+ \cap IV^+) =$$

Determining Expected Frequencies

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$$P(HIV^+) = 19/125$$

$$P(IV^+) = 48/125$$

$$P(HIV^+ \cap IV^+) = 15/125$$

If HIV and IV are independent, what should $P(HIV^+ \cap IV^+)$ equal approximately?

Determining Expected Frequencies

	HIV positive	HIV negative	Total
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$$P(HIV^+) = 19/125$$

$$P(IV^+) = 48/125$$

$$P(HIV^+ \cap IV^+) = 15/125$$

If HIV and IV are independent, what should $P(HIV^+ \cap IV^+)$ equal approximately?

$$15/125 \approx (19/125)(48/125)$$

Observed and Expected Frequencies

Observed

	HIV positive	HIV negative	Total
No IV Drug Use	4	73	77
IV Drug Use	15	33	48
Total	19	106	125

Expected

	HIV positive	HIV negative	Total
No IV Drug Use	11.704	65.296	77.000
IV Drug Use	7.296	40.704	48.000
Total	19.000	106.000	125

Observed and Expected Frequencies

Observed

	HIV positive	HIV negative	Total
No IV Drug Use	4	73	77
IV Drug Use	15	33	48
Total	19	106	125

Expected

	HIV positive	HIV negative	Total
No IV Drug Use	11.704	65.296	77.000
IV Drug Use	7.296	40.704	48.000
Total	19.000	106.000	125

$$X^2 = \sum_{i=1}^{N_1} \sum_{j=1}^{N_2} \frac{(f_O(ij) - f_E(ij))^2}{f_E(ij)} = 5.07 + 0.909 + 8.135 + 1.458 = 15.573$$

In R

```
> hiv <- cbind("positive" = c(4,15), "negative"=c(73,33))
> rownames(hiv) <- c("NoIV", "IV")

> # By hand
> N <- sum(hiv)
> status <- colSums(hiv)
> drug <- rowSums(hiv)

> exp.hiv <- diag(status*drug)/N
> exp.hiv[1,2] <- drug[1]*status[2]/N
> exp.hiv[2,1] <- drug[2]*status[1]/N

> X2 <- sum((hiv-exp.hiv)^2/exp.hiv)
> X2

[1] 15.57296

> 1-pchisq(X2,1)

[1] 0
```

Using the chisq.test Function

```
> # Using the chisq.test command  
> chisq.test(hiv, correct=FALSE)
```

Pearson's Chi-squared test

```
data:  hiv  
X-squared = 15.573, df = 1, p-value = 7.938e-05
```

Note that the chisq.test performs a continuity correction by default.

Summary of Nonparametric Tests

- ▶ Resampling techniques
- ▶ T -test alternatives
- ▶ χ^2 tests